



```

    exper |   -.412537   .0337729   -12.22   0.000   -.4787575   -.3463166
    name:  <unnamed>
    log:   C:\Users\freylejer\OneDrive - UNBC\AP_UNBC\ECON712_Winter_2026\Prob
> lem Set 3\Q2_log.smcl
    log type: smcl
    opened on: 15 Apr 2026, 16:53:17

```

```

1 .
2 . global mydirec="C:\Users\freylejer\OneDrive - UNBC\AP_UNBC\ECON712_Winter_2026
> \Problem Set 3"

3 .
4 . cd "$mydirec"
   C:\Users\freylejer\OneDrive - UNBC\AP_UNBC\ECON712_Winter_2026\Problem Set 3

5 .
6 . use "card.dta"

7 .
8 . *** PART I, i ***
9 .
10 . reg educ exper expersq black south smsa reg662 reg663 reg664 ///
>      reg665 reg666 reg667 reg668 reg669

```

Source	SS	df	MS	Number of obs	=	3,010
Model	10230.9465	13	786.995882	F(13, 2996)	=	208.09
Residual	11331.1336	2,996	3.78208732	Prob > F	=	0.0000
				R-squared	=	0.4745
				Adj R-squared	=	0.4722
Total	21562.0801	3,009	7.16586243	Root MSE	=	1.9448

educ	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
exper	-.412537	.0337729	-12.22	0.000	-.4787575	-.3463166
expersq	.0008736	.0016538	0.53	0.597	-.0023691	.0041163
black	-.9246025	.0937567	-9.86	0.000	-1.108436	-.7407685
south	-.0421616	.1352071	-0.31	0.755	-.3072698	.2229465
smsa	.5188739	.0824334	6.29	0.000	.3572421	.6805057
reg662	-.0742454	.1875067	-0.40	0.692	-.4419004	.2934097
reg663	-.0652009	.1834829	-0.36	0.722	-.4249662	.2945643
reg664	.0651931	.216739	0.30	0.764	-.3597792	.4901653
reg665	-.3522132	.2172502	-1.62	0.105	-.7781878	.0737615
reg666	-.4438211	.2333009	-1.90	0.057	-.9012672	.013625
reg667	-.3170944	.2332312	-1.36	0.174	-.7744039	.1402152
reg668	.4697064	.2676291	1.76	0.079	-.055049	.9944617
reg669	.1951717	.2027681	0.96	0.336	-.202407	.5927504
_cons	16.83694	.2355166	71.49	0.000	16.37515	17.29873

```

11 . predict d_tilde, resid
12 .

```

```

13 . * Residualise nearc4 on X_c → z_tilde
14 . reg nearc4 exper expersq black south smsa reg662 reg663 reg664 reg665 ///
> reg666 reg667 reg668 reg669

```

Source	SS	df	MS	Number of obs	=	3,010
Model	121.405116	13	9.33885506	F(13, 2996)	=	52.66
Residual	531.326113	2,996	.177345165	Prob > F	=	0.0000
				R-squared	=	0.1860
				Adj R-squared	=	0.1825
Total	652.731229	3,009	.216926298	Root MSE	=	.42112

nearc4	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
exper	-.0000383	.0073133	-0.01	0.996	-.0143779	.0143013
expersq	-2.15e-06	.0003581	-0.01	0.995	-.0007043	.0007
black	.037158	.0203023	1.83	0.067	-.00265	.0769659
south	.0203676	.0292781	0.70	0.487	-.0370396	.0777749
smsa	.3141448	.0178504	17.60	0.000	.2791446	.349145
reg662	.0118388	.0406033	0.29	0.771	-.0677742	.0914519
reg663	-.1166319	.0397319	-2.94	0.003	-.1945365	-.0387273
reg664	-.1472816	.0469333	-3.14	0.002	-.2393063	-.0552569
reg665	-.230613	.047044	-4.90	0.000	-.3228547	-.1383712
reg666	-.4115685	.0505196	-8.15	0.000	-.5106252	-.3125118
reg667	-.2987625	.0505046	-5.92	0.000	-.3977896	-.1997354
reg668	-.1698296	.0579532	-2.93	0.003	-.2834616	-.0561975
reg669	-.0506329	.043908	-1.15	0.249	-.1367258	.0354599
_cons	.6018745	.0509994	11.80	0.000	.5018771	.701872

```

15 . predict z_tilde, resid

```

```

16 .
17 . * Residualise lwage on X_c → y_tilde
18 .
19 . reg lwage exper expersq black south smsa reg662 reg663 reg664 reg665 ///
> reg666 reg667 reg668 reg669

```

Source	SS	df	MS	Number of obs	=	3,010
Model	114.030686	13	8.77159122	F(13, 2996)	=	54.91
Residual	478.610925	2,996	.159749975	Prob > F	=	0.0000
				R-squared	=	0.1924
				Adj R-squared	=	0.1889
Total	592.641611	3,009	.196956335	Root MSE	=	.39969

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
exper	.0540272	.006941	7.78	0.000	.0404175	.0676368
expersq	-.0022159	.0003399	-6.52	0.000	-.0028823	-.0015494
black	-.2690631	.0192689	-13.96	0.000	-.3068447	-.2312815
south	-.1480811	.0277878	-5.33	0.000	-.2025662	-.093596
smsa	.1918217	.0169417	11.32	0.000	.1586031	.2250403
reg662	.0914433	.0385364	2.37	0.018	.0158827	.1670038
reg663	.1397198	.0377095	3.71	0.000	.0657808	.2136589
reg664	.0549257	.0445442	1.23	0.218	-.0324147	.1422662
reg665	.095719	.0446493	2.14	0.032	.0081726	.1832654
reg666	.0977417	.0479481	2.04	0.042	.0037273	.1917562
reg667	.0894529	.0479337	1.87	0.062	-.0045335	.1834393
reg668	-.0212046	.0550032	-0.39	0.700	-.1290525	.0866432
reg669	.1342786	.0416729	3.22	0.001	.0525682	.2159891
_cons	5.884744	.0484034	121.58	0.000	5.789837	5.979652

20 . predict y_tilde, resid

21 .

22 .

23 . *** PART I, ii ***

24 .

25 . reg d_tilde z_tilde

Source	SS	df	MS	Number of obs	=	3,010
Model	56.4529005	1	56.4529005	F(1, 3008)	=	15.06
Residual	11274.6807	3,008	3.74823161	Prob > F	=	0.0001
				R-squared	=	0.0050
				Adj R-squared	=	0.0047
Total	11331.1336	3,009	3.76574729	Root MSE	=	1.936

d_tilde	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
z_tilde	.3259587	.083991	3.88	0.000	.1612731	.4906442
_cons	2.19e-09	.0352882	0.00	1.000	-.0691915	.0691915

26 . local b_fs = string(_b[z_tilde], "%6.4f")

27 .

28 . reg y_tilde z_tilde

Source	SS	df	MS	Number of obs	=	3,010
Model	1.18731521	1	1.18731521	F(1, 3008)	=	7.48
Residual	477.423609	3,008	.158717955	Prob > F	=	0.0063
				R-squared	=	0.0025
				Adj R-squared	=	0.0021
Total	478.610924	3,009	.159059795	Root MSE	=	.39839

y_tilde	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
z_tilde	.0472718	.0172835	2.74	0.006	.0133831	.0811606
_cons	1.49e-11	.0072616	0.00	1.000	-.0142381	.0142381

29 . local b_rf = string(_b[z_tilde], "%6.4f")

30 .

```
31 . twoway (scatter d_tilde z_tilde, msize(tiny) mcolor(gs10)) ///
>         (lfit d_tilde z_tilde, lcolor(black) lwidth(medthick)), ///
>         xtitle("Residualised nearc4 ") ///
>         ytitle("Residualised education ") ///
>         title("First Stage (FWL)") ///
>         note("Slope = `b_fs'", size(small)) ///
>         legend(off)
```

32 . quietly graph export "PS3_Q2a_FirstStage.png", replace width(1200)

```

33 .
34 . * Reduced-form scatter: y_tilde on z_tilde
35 . twoway (scatter y_tilde z_tilde, msize(tiny) mcolor(gs10)) ///
>         (lfit y_tilde z_tilde, lcolor(black) lwidth(medthick)), ///
>         xtitle("Residualised nearc4 ")          ///
>         ytitle("Residualised log wage ")        ///
>         title("Reduced Form (FWL)")              ///
>         note("Slope = `b_rf'", size(small))      ///
>         legend(off)

36 . quietly graph export "PS3_Q2a_ReducedForm.png", replace width(1200)

37 .
38 .
39 . *** PART I, iii ***
40 .
41 . reg educ nearc4 exper expersq black south smsa reg662 reg663 reg664 reg665 ///
>         reg666 reg667 reg668 reg669

```

Source	SS	df	MS	Number of obs	=	3,010
Model	10287.3994	14	734.81424	F(14, 2995)	=	195.20
Residual	11274.6807	2,995	3.76450107	Prob > F	=	0.0000
				R-squared	=	0.4771
				Adj R-squared	=	0.4747
Total	21562.0801	3,009	7.16586243	Root MSE	=	1.9402

educ	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
nearc4	.3259587	.0841731	3.87	0.000	.1609158	.4910015
exper	-.4125245	.0336943	-12.24	0.000	-.4785909	-.3464582
expersq	.0008743	.0016499	0.53	0.596	-.0023608	.0041095
black	-.9367145	.0935907	-10.01	0.000	-1.120223	-.7532059
south	-.0488006	.1349033	-0.36	0.718	-.3133131	.2157118
smsa	.4164757	.0863879	4.82	0.000	.24709	.5858614
reg662	-.0781043	.187073	-0.42	0.676	-.4449088	.2887002
reg663	-.0271837	.1833189	-0.15	0.882	-.3866274	.33226
reg664	.1132008	.2165896	0.52	0.601	-.3114786	.5378801
reg665	-.2770429	.217612	-1.27	0.203	-.703727	.1496413
reg666	-.3096667	.2353218	-1.32	0.188	-.7710754	.1517419
reg667	-.2197101	.2340433	-0.94	0.348	-.6786121	.2391918
reg668	.5250638	.2673885	1.96	0.050	.00078	1.049348
reg669	.2116759	.202341	1.05	0.296	-.1850654	.6084173
_cons	16.64075	.2403679	69.23	0.000	16.16945	17.11206

```

42 . test nearc4

```

```

( 1) nearc4 = 0

```

```

      F( 1, 2995) = 15.00
      Prob > F = 0.0001

```

```
43 . display "F-stat on nearc4 = " %6.2f r(F) " (p = " %5.3f r(p) ")"
    F-stat on nearc4 = 15.00 (p = 0.000)
```

```
44 .
45 .
46 . ***** PART II *****
47 .
48 .
49 . reg lwage educ exper black south married smsa, robust
```

```
Linear regression               Number of obs   =    3,003
                               F(6, 2996)       =    230.88
                               Prob > F          =    0.0000
                               R-squared         =    0.3050
                               Root MSE      =    .37025
```

lwage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
educ	.0711729	.0036137	19.70	0.000	.0640872	.0782585
exper	.0341518	.0022668	15.07	0.000	.0297072	.0385964
black	-.1660274	.0173977	-9.54	0.000	-.2001402	-.1319147
south	-.1315518	.0152007	-8.65	0.000	-.1613567	-.1017469
married	-.0358707	.0035727	-10.04	0.000	-.042876	-.0288655
smsa	.1757871	.0150994	11.64	0.000	.1461808	.2053934
_cons	5.063317	.0661786	76.51	0.000	4.933557	5.193077

```
50 . eststo OLS
```

```
51 .
52 .
53 . ivreg2 lwage (educ = nearc4) exper black south married smsa, first ffirst ro
    > bust
```

First-stage regressions

First-stage regression of educ:

Statistics robust to heteroskedasticity
Number of obs = 3003

educ	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
nearc4	.3272826	.0805341	4.06	0.000	.1693748	.4851903
exper	-.404434	.0081984	-49.33	0.000	-.420509	-.388359
black	-.9475281	.0884964	-10.71	0.000	-1.121048	-.7740082
south	-.2973528	.0787506	-3.78	0.000	-.4517635	-.1429421
married	-.0726936	.0176169	-4.13	0.000	-.107236	-.0381512
smsa	.4208945	.0848857	4.96	0.000	.2544543	.5873348
_cons	16.8307	.1252943	134.33	0.000	16.58502	17.07637

F test of excluded instruments:

```
F( 1, 2996) = 16.52
Prob > F     = 0.0000
```

Sanderson-Windmeijer multivariate F test of excluded instruments:

```
F( 1, 2996) = 16.52
Prob > F     = 0.0000
```

Summary results for first-stage regressions

Variable	F(1, 2996)	P-val	(Underid)		P-val	(Weak id)	
			SW	Chi-sq(1)		SW	F(1, 2996)
educ	16.52	0.0000		16.55	0.0000		16.52

NB: first-stage test statistics heteroskedasticity-robust

Stock-Yogo weak ID F test critical values for single endogenous regressor:

10% maximal IV size	16.38
15% maximal IV size	8.96
20% maximal IV size	6.66
25% maximal IV size	5.53

Source: Stock-Yogo (2005). Reproduced by permission.

NB: Critical values are for i.i.d. errors only.

Underidentification test

Ho: matrix of reduced form coefficients has rank=K1-1 (underidentified)

Ha: matrix has rank=K1 (identified)

Kleibergen-Paap rk LM statistic Chi-sq(1)=16.36 P-val=0.0001

Weak identification test

Ho: equation is weakly identified

Cragg-Donald Wald F statistic 15.77

Kleibergen-Paap Wald rk F statistic 16.52

Stock-Yogo weak ID test critical values for K1=1 and L1=1:

10% maximal IV size	16.38
15% maximal IV size	8.96
20% maximal IV size	6.66
25% maximal IV size	5.53

Source: Stock-Yogo (2005). Reproduced by permission.

NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.

Weak-instrument-robust inference

Tests of joint significance of endogenous regressors B1 in main equation

Ho: B1=0 and orthogonality conditions are valid

Anderson-Rubin Wald test F(1,2996)= 6.34 P-val=0.0119

Anderson-Rubin Wald test Chi-sq(1)= 6.35 P-val=0.0117

Stock-Wright LM S statistic Chi-sq(1)= 6.36 P-val=0.0117

NB: Underidentification, weak identification and weak-identification-robust test statistics heteroskedasticity-robust

Number of observations	N =	3003
Number of regressors	K =	7
Number of endogenous regressors	K1 =	1
Number of instruments	L =	7
Number of excluded instruments	L1 =	1

IV (2SLS) estimation

Estimates efficient for homoskedasticity only

Statistics robust to heteroskedasticity

		Number of obs =	3003
		F(6, 2996) =	148.44
		Prob > F =	0.0000
Total (centered) SS	=	Centered R2 =	0.2513
Total (uncentered) SS	=	Uncentered R2 =	0.9963
Residual SS	=	Root MSE =	.3838

lwage	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
educ	.1241642	.0491577	2.53	0.012	.0278169	.2205116
exper	.0555882	.0198674	2.80	0.005	.0166488	.0945277
black	-.1156855	.0495591	-2.33	0.020	-.2128197	-.0185514
south	-.1131647	.0229476	-4.93	0.000	-.1581411	-.0681883
married	-.0319754	.0050725	-6.30	0.000	-.0419172	-.0220335
smsa	.1477065	.0303171	4.87	0.000	.0882861	.2071269
_cons	4.162476	.8348751	4.99	0.000	2.526151	5.798801

Underidentification test (Kleibergen-Paap rk LM statistic): **16.356**
Chi-sq(1) P-val = **0.0001**

Weak identification test (Cragg-Donald Wald F statistic): **15.767**
(Kleibergen-Paap rk Wald F statistic): **16.515**
Stock-Yogo weak ID test critical values: 10% maximal IV size **16.38**
15% maximal IV size **8.96**
20% maximal IV size **6.66**
25% maximal IV size **5.53**

Source: Stock-Yogo (2005). Reproduced by permission.

NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.

Hansen J statistic (overidentification test of all instruments): **0.000**
(equation exactly identified)

Instrumented: educ
Included instruments: exper black south married smsa
Excluded instruments: nearc4

54 . eststo tsls

55 . matrix first=e(first)

56 . estadd scalar F1=first[4,1]

added scalar:

$$e(F1) = 16.515286$$

57 .

58 .

59 .

```
60 . estout OLS tsls using "PS3_Q2b_Card_Table57.tex",          ///
>   replace                                                    ///
>   style(tex)                                                  ///
>   cells(b(star fmt(%9.4f)) se(par fmt(%9.4f)))              ///
>   starlevels(* 0.10 ** 0.05 *** 0.01)                       ///
>   stats(F1 N r2,                                             ///
>     labels("F-test" "Observations" "\$R^2\$")              //
> /
>   fmt(%9.0f %9.0f %9.3f))                                     ///
>   mlabels("OLS" "2SLS")                                       ///
>   collabels(none)                                             ///
>   keep(educ exper black south married smsa)                  //
> /
>   prehead(                                                    ///
>     "\begin{table}[H]"                                         ///
>     "\begin{center}"                                          ///
>     "\caption{Replication Table 57}"                          ///
>   ///
>     "\resizebox{.57\textwidth}{!}{%"                        ///
>     "\begin{tabular}{lll}"                                    ///
>     "\hline" "\hline"                                         ///
```

```

>      )                                     ///
>      postfoot(                             ///
>      "\hline" "\hline"                     ///
>      "\end{tabular}}"                       ///
>      "\end{center}"                         ///
>      "\captionsetup{font={scriptsize}}"     ///
>      "\captionsetup{width=0.80\textwidth}"  ///
>      "\captionsetup{skip=0pt}"              ///
>      "\caption*{\scriptsize Notes: Heteroscedasticity-robust standard error
> s in parentheses." ///
>      "Controls: region dummies, and a constant." ///
>      "Instrument in 2SLS: \texttt{nearc4} (proximity to 4-year college)." /
> //
>      "* \p<0.10\$, ** \p<0.05\$, *** \p<0.01\$." ///
>      "\end{table}"                          ///
>      )
(output written to PS3_Q2b Card Table57.tex)

```

```

61 .
62 . cap log close

```